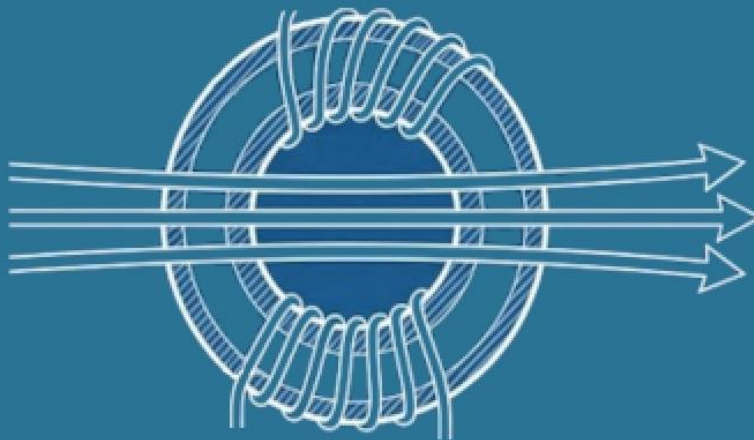


Behavioral modeling of a Fluxgate current sensor in LTspice

From electromagnetism to circuit
simulation



Leonardo Chieco

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Fluxgate current sensors are among the most accurate technologies available for measuring direct currents and weak magnetic fields. Their operating principle is based on the nonlinear magnetization of a ferromagnetic core, which is driven periodically into saturation, and on the synchronous demodulation of the induced signal, making it possible to accurately determine the magnitude of the external magnetic field or the DC current being measured.

This tutorial was created with the goal of providing a thorough understanding of the operating principle of the Fluxgate sensor through a simple yet physically consistent behavioral model implemented in LTspice. Starting from the fundamental laws of electromagnetism, we will progressively build a behavioral model of the sensor by introducing the nonlinear $B(H)$ characteristic, magnetic flux generation, the voltage induced in the pickup winding, and the synchronous demodulation of the output signal. The objective is not to develop a complex electromagnetic model, but rather to provide an educational tool that bridges the gap between theoretical concepts and circuit simulation, helping readers understand why the Fluxgate sensor is capable of measuring DC currents with exceptionally high sensitivity.

Enjoy the reading!

About the Author

Leonardo Chieco is an electronic engineer with more than 20 years of experience in the design and development of automation control software (PC/PLC), industrial electronic hardware, embedded firmware, robotics, and mechatronic systems.

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1. The Fluxgate sensor

The operating principle of a fluxgate sensor is based on the fundamental laws of electromagnetism and, in particular, on the relationship between electric current, magnetic field, magnetic flux, and induced voltage. The first fundamental law is Ampère's law, which states that an electric current flowing through a winding generates a magnetic field inside the ferromagnetic core. In the case of a closed magnetic circuit with an average magnetic path length l_m , the magnetizing field can be expressed as

$$H = \frac{N I}{l_m}$$

where N represents the number of turns in the winding and I is the current flowing through it. This relationship shows that the magnetic field is directly proportional to the number of ampere-turns applied to the core and constitutes the starting point for describing the behavior of the fluxgate sensor.

The magnetizing field produces a magnetic flux density B inside the ferromagnetic material, whose general constitutive relationship is

$$B = \mu H$$

where μ represents the magnetic permeability of the material. Unlike linear materials, in ferromagnetic materials the permeability is not constant but depends on the applied magnetic field because magnetization results from the progressive alignment of magnetic domains. For low values of H , most domains are still randomly oriented, and small increases in the magnetic field produce large changes in magnetization; consequently, the permeability assumes very high values. As the magnetic field increases, more and more domains align with the field direction until virtually all of them point in the same direction. Under these conditions, the material reaches magnetic saturation, and further increases in the magnetic field produce only a slight increase in magnetic flux density, mainly due to the permeability of free space. As a result, the $B(H)$ characteristic exhibits a strongly nonlinear behavior, initially almost linear and then gradually saturating.

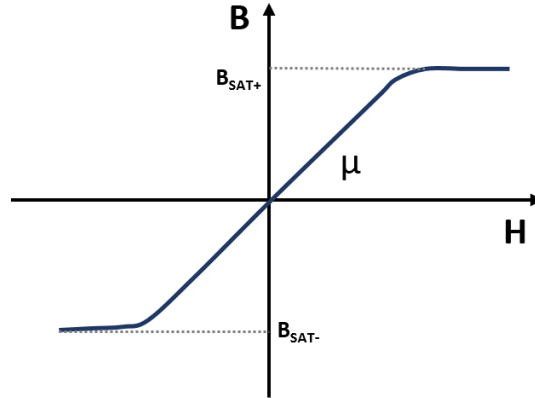


Figure 1: Magnetic flux density B as a function of magnetic field strength H

A simplified yet highly effective representation of this behavior is provided by the hyperbolic tangent function:

$$B(H) = B_{SAT} \tanh\left(\frac{H}{H_{SAT}}\right)$$

This choice is not arbitrary, since the hyperbolic tangent qualitatively reproduces the transition between the nearly linear region and magnetic saturation while maintaining continuity and differentiability, both of which are fundamental characteristics for LTspice simulations.

To give physical meaning to these parameters, let us consider a real ferrite or amorphous alloy core with an effective cross-sectional area A_E of approximately 100 mm², corresponding to 100×10⁻⁶ m², and an average magnetic path length L_m of about 10 cm. If the material has a typical saturation flux density B_{SAT} of approximately 0.3–0.4 Tesla, a realistic value is 0.35 T. The parameter H_{SAT} represents the magnetic field strength at which the material enters saturation; for many ferrite materials, this value ranges between 100 and 200 A/m, making 150 A/m a realistic choice. If we consider an excitation winding with 50 turns, an alternating current of a few tens of milliamperes generates a magnetic field H of the appropriate magnitude to cyclically drive the core close to saturation:

$$H = \frac{N I}{l_m} = \frac{50 \cdot 30mA}{10 \text{ cm}} = \frac{50 \cdot 3 \cdot 10^{-2} A}{10^{-2} \text{ m}} = 150 \frac{A}{m}$$

The operating principle of the fluxgate sensor consists precisely in driving the magnetic core cyclically into saturation by means of an excitation winding supplied with a sinusoidal current. According to Ampère's law, the magnetizing field generated by the excitation winding is

$$H_{exc} = \frac{N_1 I_{exc}}{l_m}$$

which is superimposed on the magnetic field generated by the external current to be measured

$$H_{ext} = \frac{N_{measure} I_{measure}}{l_m}$$

The total magnetic field therefore becomes

$$H_{tot} = H_{exc} + H_{ext}$$

If the external magnetic field is zero, the core follows a perfectly symmetrical magnetization curve during both the positive and negative half-cycles of the excitation signal. Consequently, the magnetic response is symmetrical, and the magnetic flux contains primarily the fundamental component and odd harmonics.

When a DC magnetic field is present, however, the operating point of the material shifts along the B(H) curve, breaking the symmetry of the saturation process. The core reaches saturation earlier during one half-cycle and later during the other, producing a distortion of the magnetic flux waveform and generating even harmonics, particularly the second harmonic, whose amplitude is directly proportional to the applied external magnetic field.

The magnetic flux in the core is given by

$$\Phi = B(H) A_E$$

where A_E is the cross-sectional area of the magnetic circuit.

According to the Faraday–Neumann–Lenz law, the time variation of the magnetic flux linked with a winding induces an electromotive force

$$V = -N_{pick} \frac{d\Phi}{dt}$$

therefore, the secondary winding of the fluxgate provides a voltage proportional to the time derivative of the magnetic flux.

Once the physical model has been understood, the SPICE netlist represents the circuit-level implementation of these concepts.

The first block defines the magnetic geometry and the material parameters, which constitute the "constitutive description" of the magnetic core.

```
* --- CORE GEOMETRY & MAGNETIC PARAMETERS ---
* Nexc: Number of turns of the excitation winding [dimensionless]
.param Nexc=50
* Npick: Number of turns of the pickup/sense winding [dimensionless]
.param Npick=100
* Nmeas: Number of turns of the measurement/primary circuit [dimensionless]
.param Nmeas=1
* Ae: Effective cross-sectional area of the magnetic core [m^2]
.param Ae=100u
* Le: Mean magnetic path length of the core [meters, m]
.param Le=0.1
* Bsat: Saturation magnetic flux density of the core material [Tesla, T]
.param Bsat=0.35
* Hsat: Saturation magnetic field strength intensity [Ampere/meter, A/m]
.param Hsat=150
* R_excitation: Series resistance of the excitation circuit [Ohm]
.param R_excitation=10

* --- PRIMARY CIRCUIT (CURRENT TO BE MEASURED) ---
* I_measure: The DC input current passing through the sensor [Ampere, A].
.param I_measure=2.0
```

The V_{EXT} source operating at 20 kHz represents the periodic excitation that forces the magnetic hysteresis cycle, while the series resistance models the losses and impedance of the real excitation winding.

```
* --- AC EXCITATION CIRCUIT (20 kHz) ---
Vexc exc 0 SINE(0 10 20k)
Rexc exc 0 {R_excitation}
```

The key element is the behavioral source Bphi, which directly implements the equation

$$\Phi = A_E B_{SAT} \tanh\left(\frac{H}{H_{SAT}}\right)$$

where H is obtained from the sum of the AC contribution generated by the excitation winding and the DC contribution produced by the primary current. This block effectively replaces the entire physical magnetic behavior with a continuous analytical model while preserving the nonlinearity.

```
* --- MAGNETIC BEHAVIORAL CORE EQUATIONS ---
* Bphi: Calculates the total flux [Weber, Wb] based on Htot (1V = 1Wb)
Bphi phi 0 V = {Ae} * {Bsat} * tanh((( {Nexc}*(V(exc)/{R_excitation}) + ({Nmeas}*{I_measure})
) / {Le}) / {Hsat})
```

The second step converts the magnetic flux into voltage through E_{pick} , which implements Faraday's law in discrete form by using the time derivative of the magnetic flux signal. This stage is essential because it transforms a magnetic physical quantity into an observable electrical signal.

```
* --- PICKUP / SENSE CIRCUIT (OUTPUT) ---
* Faraday's Law of Induction: V_induced = Npick * d(Phi)/dt [Volts, V]
Epick out_raw 0 VALUE = { {Npick} * ddt(V(phi)) }

* Attenuation network to scale ideal spikes down below +/-10V for the IC
Eattenuate out 0 VALUE = { V(out_raw) / 10 }
Rload out 0 1k
```

The most conceptually significant part is the synchronous demodulator implemented by the Bmultiplier block.

In a real fluxgate system, the measurement is not extracted directly from the secondary winding signal but through correlation with a sinusoidal reference

operating at twice the excitation frequency. This is because the useful information is contained in the second harmonic, whose amplitude is proportional to the asymmetry introduced by the primary current. Multiplying the measured signal by the reference performs an operation equivalent to an analog lock-in amplifier, converting a high-frequency AC variation into a DC component proportional to the measured quantity.

The following figure shows the symmetrical magnetic flux waveform when the current to be measured is zero. Under these conditions, no second harmonic is present in the frequency spectrum.

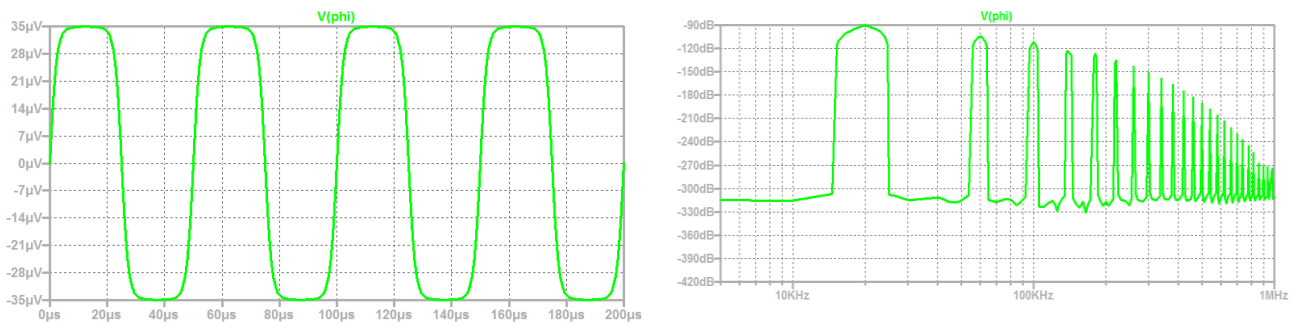


Figure 2: Magnetic flux and FFT with zero measured current

The following figure, on the other hand, shows the asymmetrical magnetic flux waveform when the measured current is 10 A. In this case, the second harmonic appears clearly in the frequency spectrum at 40 kHz.

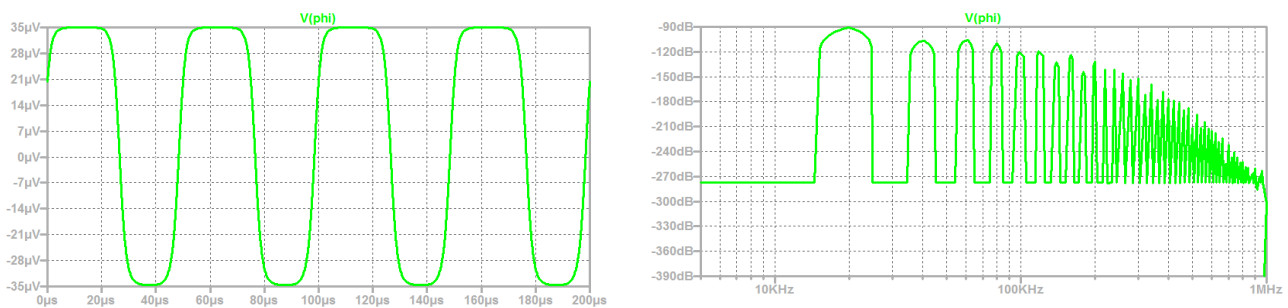


Figure 3: Asymmetrical magnetic flux and FFT with a measured current of 10 A

Let us examine in detail how synchronous demodulation works.

Suppose the sensed voltage is:

$$v_s(t) = A_1 \sin(\omega t) + A_2 \sin(2\omega t) + A_3 \sin(3\omega t) + \dots$$

while the reference signal is

$$v_r(t) = \sin(2\omega t)$$

The multiplier produces:

$$v_m(t) = v_s(t) v_r(t) = [A_1 \sin(\omega t) + A_2 \sin(2\omega t) + A_3 \sin(3\omega t) + \dots] \sin(2\omega t)$$

Recall the trigonometric identity:

$$\sin(\alpha) \sin(\beta) = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

Expanding each term:

- Fundamental component:

$$A_1 \sin(\omega t) \sin(2\omega t) = \frac{A_1}{2} [\cos(\omega t) - \cos(3\omega t)]$$

No DC component appears.

- Second harmonic:

$$A_2 \sin(2\omega t) \sin(2\omega t) = \frac{A_2}{2} [\cos(0) - \cos(4\omega t)] = \frac{A_2}{2} - \frac{A_2}{2} \cos(4\omega t)$$

Here a DC component equal to $\frac{A_2}{2}$ appears.

- Third harmonic:

$$A_3 \sin(3\omega t) \sin(2\omega t) = \frac{A_3}{2} [\cos(\omega t) - \cos(5\omega t)]$$

Again, no DC component is produced.

Therefore, $v_m(t)$ contains a DC component equal to $\frac{A_2}{2}$ together with several harmonic components.

```
* --- SYNCHRONOUS DEMODULATOR STAGE ---
* Vref: Reference voltage source operating at 2f (40 kHz), 1V Peak amplitude [Volts, V]
Vref ref 0 SINE(0 1 40k)

* Bmultiplier: Models the transfer function: W = (X * Y) / 10V
* Integrated with a hard-limit clipper at +/-13.5V to simulate rail saturation (+/-15V supply)
Bmultiplier v_mult 0 V = limit( (V(out) * V(ref)) / 10, -13.5, 13.5 )
```

The low-pass filter formed by R_f and C_f removes all remaining AC components, leaving only the DC component that represents the reconstructed value of the primary current. In this way, the entire signal chain behaves as a transducer: a DC electrical quantity is converted into a nonlinear magnetic modulation, then into a complex AC signal, and finally back into a DC voltage proportional to the measured current.

```
* --- LOW-PASS FILTER (LPF) ---
* Cutoff frequency: f_c = 1 / (2 * pi * Rf * Cf) ~= 159 Hz
Rf v_mult v_dc 10k
Cf v_dc 0 100n

* --- MONITORING NODES FOR B-H GRAPH PLOTTING ---
Bh h_test 0 V = ( {Nexc}*(V(exc)/{R_excitation}) + ({Nmeas}*{I_measure}) ) / {Le}
Bb b_test 0 V = {Bsat} * tanh(V(h_test) / {Hsat})
```

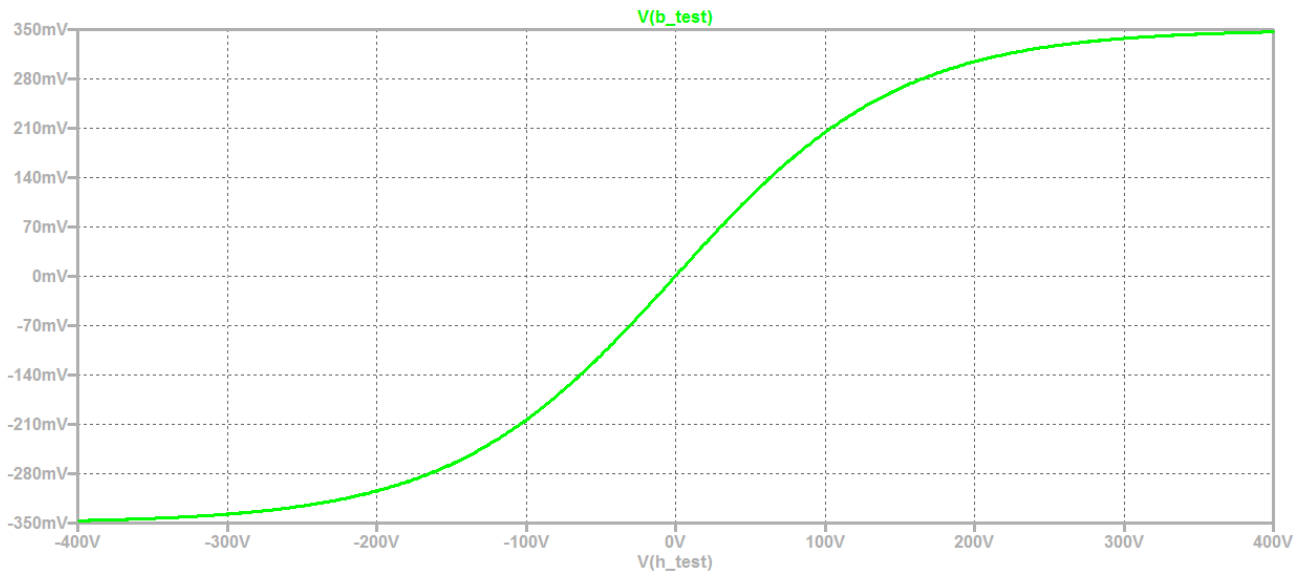


Figure 4: B(H) characteristic curve of the selected core

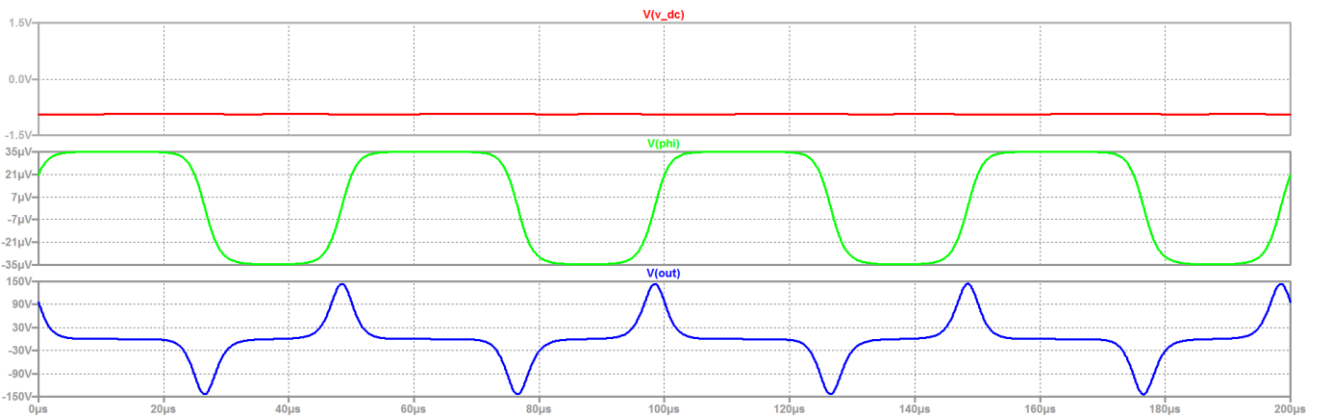


Figure 5: Waveforms and output signal with a 10 A input current

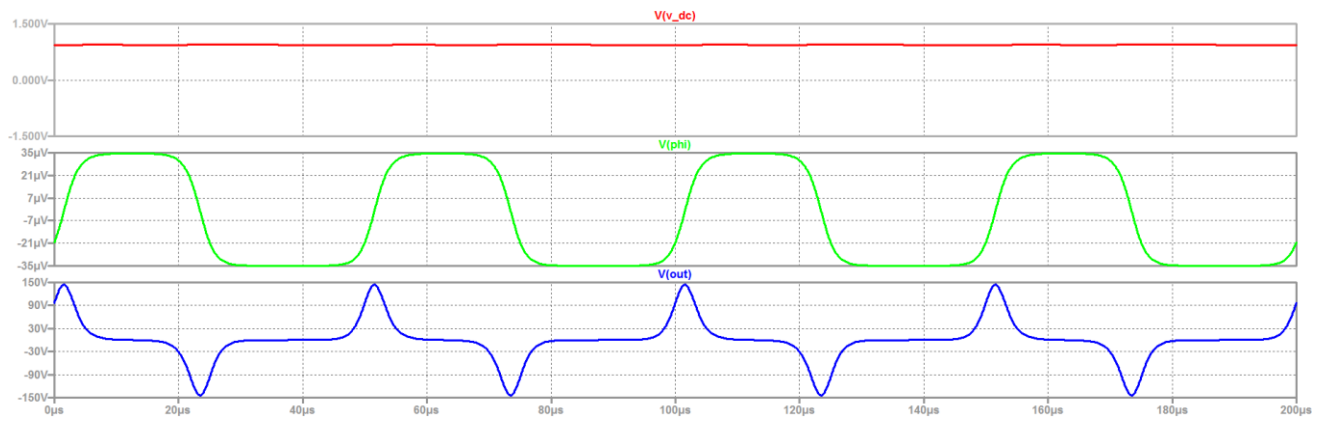


Figure 6: Waveforms and output signal with a -10 A input current

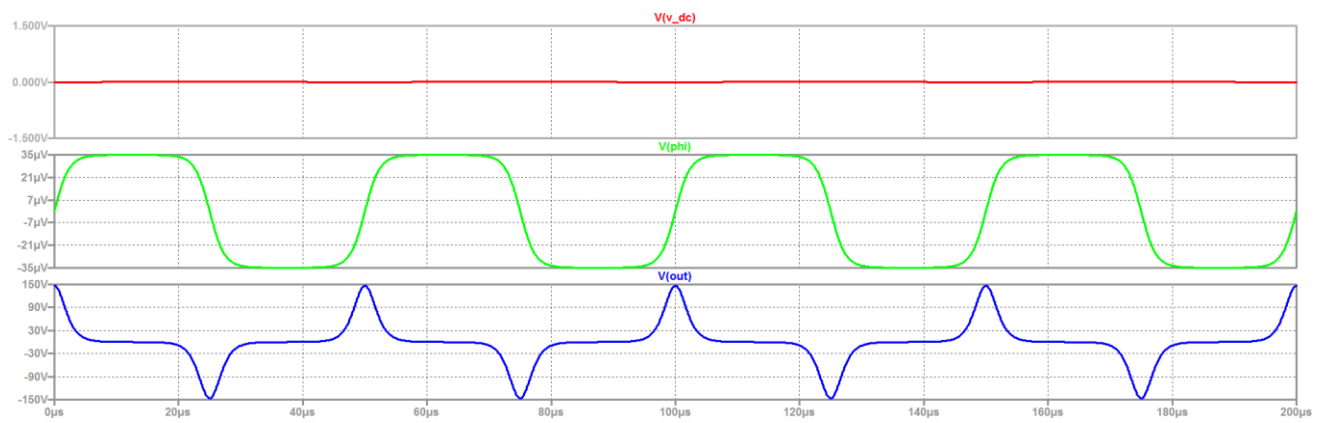


Figure 7: Waveforms and output signal with no input current